Gauss

Building Models to Describe our World
Gauss's Law, Flux and Application

Outline

Gauss’s Law and Flux

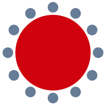
Building Models to Describe our World

Gauss’s Law, Flux and Application

Gauss's Law

$$\boxed{\Phi_E = \frac{Q^{enc}}{\epsilon_0}}$$

An analogy to understand flux through a closed surface

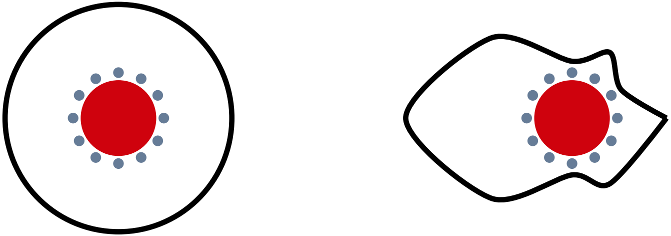


Pellets being emitted isotropically by a source.

- A “source” emits little pellets isotropically (with spherical symmetry).
- Every second, the source emits 12 pellets.

An analogy to understand flux through a closed surface

- One can think of flux as the number of pellets that are coming out of the source and crossing the closed surface.
- The flux through the closed surface does not depend on the surface, it only depends on the source!



Pellets being emitted isotropically by a source.

$1/R^2$

- The flux of pellets per unit area through a spherical surface decreases as:

$$N(R) \propto \frac{1}{4\pi R^2}$$

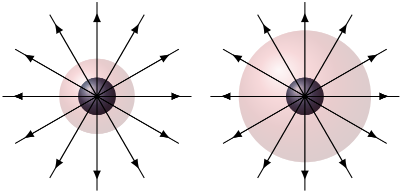
- This gives us a sense of how the electric force (or any $1/R^2$ force) is actually transmitted:
    - Rather than action at a distance, maybe it's more like particles being emitted from a charge and the force being related to the density of particles that is being felt (the particles are photons for the electric force).
    - We had already argued that waves also have intensities that decrease as $1/R^2$, so maybe waves transmit these forces (light waves transmit the electric force)
    - Wait, what? (photons are particles of light and light waves, it's weird!)

  $\rightarrow$ Gravity waves (confirmed), gravitons (particles for gravity, one of the major problems in physics!)

Gauss' Law

$$\Phi_E = \frac{Q^{enc}}{\epsilon_0}$$

- The flux of a field (e.g. Electric Field) across a **closed surface** only depends on the source (e.g. Charge) enclosed by that surface.
- Think of flux through a surface as the total number of "pellets' hitting the surface or the number of field lines crossing the surface.



The number of field lines crossing the surface does not depend on the size of the surface, it only depends on the charge that is enclosed.

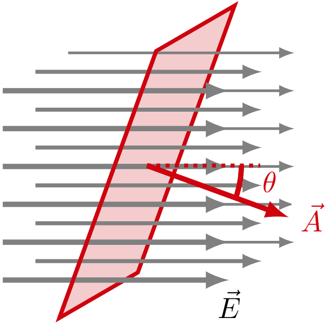
Calculating Flux through a surface

Flux is a general quantity that can be calculated given a vector field and a surface. Think of it as a measure of the number of field lines crossing the surface. Given:

- a vector field, $\vec{E}$.
- a surface, represented by a vector, $\vec{A}$:
  - The **magnitude of** $\vec{A}$ is the area of the surface.
  - The **direction of** $\vec{A}$ is perpendicular to the surface (parallel to the "normal" vector of the surface). There are two choices.

The flux of the field through that surface is defined by:

$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

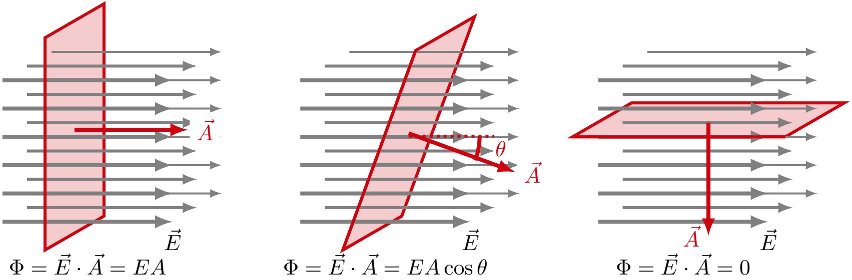


Flux through an open surface.

Flux through a surface

$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

- This definition of flux meets the requirements for the analogy with pellets or field lines:
  - If the surface is perpendicular to the field, the most field lines cross the surface.
  - If the surface is parallel to the field, no field lines cross the surface.
  - If the electric field is stronger, more field lines cross the surface.
- **For a closed surface, we define the direction of $\vec{A}$ to be outwards.**

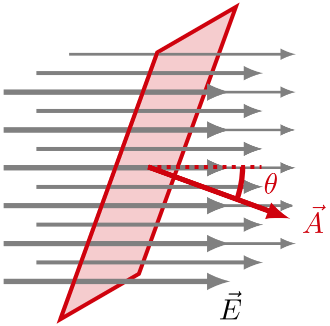


Flux through an open surface.

Flux through a surface and scalar product

$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

- The scalar product “selects” the part of the surface vector that is parallel to the field: $A_\perp = A\cos\theta$.
- The scalar product “selects” the part of the field that is parallel to the surface vector: $E_\perp = E\cos\theta$.

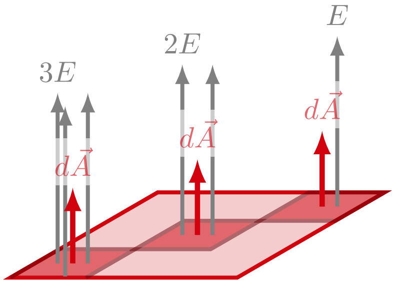


Flux through an open surface. $A\cos\theta$ is the part of $\vec{A}$ that is parallel to $\vec{E}$.

What if the field changes across the surface?

- If the field is different across each point on the surface, need to break the surface up into smaller surface elements, where the field is constant:
    - The direction of $\vec{E}$ can change with respect to the surface normal.
    - The magnitude of $\vec{E}$ can change.
- **The flux through the total surface is the sum of the fluxes through each surface element**:

$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$

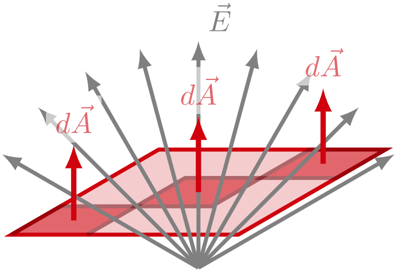


Dividing a surface into smaller elements when the $\vec{E} \cdot \vec{A}$ changes across the surface.

What if the field changes across the surface?

- If the field is different across each point on the surface, need to break the surface up into smaller surface elements, where the field is constant:
    - The direction of $\vec{E}$ can change with respect to the surface normal.
    - The magnitude of $\vec{E}$ can change.
- **The flux through the total surface is the sum of the fluxes through each surface element**:

$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$



Dividing a surface into smaller elements when the $\vec{E} \cdot \vec{A}$ changes across the surface.

More general definition of flux

- In general, if the field varies continuously over the surface S, we need to **take infinitessimally small area elements, $d\vec{A}$,** and sum (integrate) the infinitely many fluxes from each area element.
- The flux becomes (a 2D "surface integral"):

$$\Phi_E = \int_S \vec{E} \cdot d\vec{A}$$

- **Think of adding together a bunch of little fluxes**, $d\Phi$:

$$d\Phi = \vec{E} \cdot d\vec{A}$$

- The subscript S indicates that we are integrating over the 2D surface, S. In general, this is a "double integral" and rather painful to solve, *so we want to avoid having to actually do the integral!*

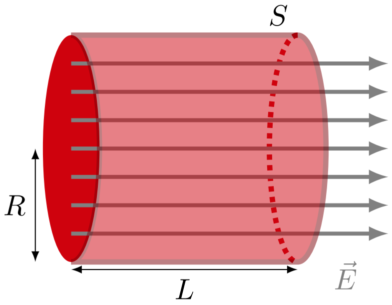
$$\int_S \vec{E} \cdot d\vec{A} = \int \int E(x, y) \cos \theta(x, y) dx dy$$

Strategy for the flux integral

- Remember, the integral is a sum, so we can break the integral over the whole surface into integrals over more convenient surfaces:

$$\Phi_E = \int_S \vec{E} \cdot d\vec{A} = \int_{S_1} \vec{E} \cdot d\vec{A} + \int_{S_2} \vec{E} \cdot d\vec{A}$$

- **We want to avoid doing an actual integral!**
- We always try to break the surface up into sub surfaces over which the electric field is constant.
- When we use Gauss' Law, we can choose the surface, so we choose one where the integrals are easy!



Flux through a closed cylinder.

Application of Gauss’s Law

Building Models to Describe our World
Gauss’s Law, Flux and Application

Using Gauss' Law

- Only practical to use when we can actually calculate the flux (do the integral):

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \oint_S E \cos\theta dA$$

- This means **choosing a clever surface**:
  - One that goes through a point where we want to know the electric field (kinda the point).
  - One where $\vec{E}$ is constant, ideally, or,
  - One where $\vec{E}$ is constant in magnitude.
  - One where we can calculate its area easily:

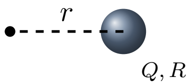
$$\Phi_E = E \cos\theta \oint_S dA = E \cos\theta A$$

- The integral is only ever easy if the problem has a high degree of symmetry.

Field outside of sphere of radius, R, carrying charge Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre: Need to **choose a Gaussian surface** that has:

- $\vec{E}$ always parallel or perpendicular to it.
- $\vec{E}$ constant in magnitude.

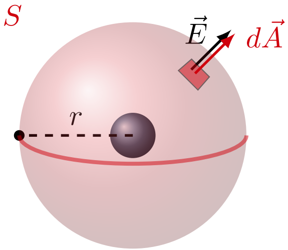


Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field outside of sphere of radius, R, carrying charge Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.



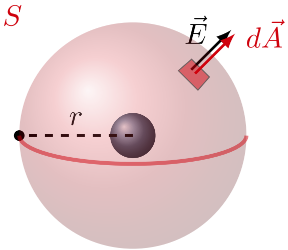
Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field outside of sphere of radius, R, carrying charge Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral (use symmetry arguments):

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \oint E(r)dA = E(r) \oint dA$$



Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field outside of sphere of radius, R, carrying charge Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

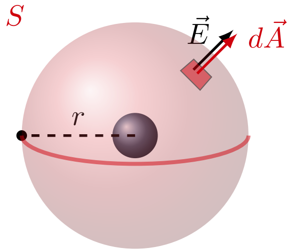
1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral (use symmetry arguments):

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \oint E(r) dA = E(r) \oint dA$$

→ Adding together all of the dA:

$$\oint dA = 4\pi r^2$$

$$\therefore \oint \vec{E}(\vec{r}) \cdot d\vec{A} = 4\pi r^2 E(r)$$



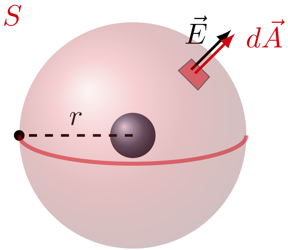
Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field outside of sphere of radius, R, carrying charge Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $4\pi r^2 E(r)$
3. Determine the charge enclosed. In this case, all of Q is enclosed:

$$Q^{enc} = Q$$



Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field outside of sphere of radius, R, carrying charge Q

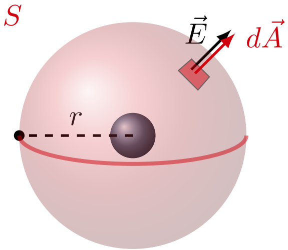
Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $-4\pi r^2 E(r)$
3. Determine the charge enclosed: $Q^{enc} = Q$
4. Put it all together:

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

$$4\pi r^2 E(r) = \frac{Q}{\epsilon_0}$$

$$\therefore \boxed{E(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}}$$



Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Steps to Using Gauss' Law

$$\Phi_E = \frac{Q^{enc}}{\epsilon_0}$$

Three steps:

1. Calcuate the flux.
2. Calculate the charge enclosed.
3. Put it all together.

Using Gauss' Law

- Only practical to use when we can actually calculate the flux (do the integral):

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \oint_S E \cos\theta dA$$

- This means **choosing a clever surface**:
    - One that goes through a point where we want to know the electric field (kinda the point).
    - One where $\vec{E}$ is constant, ideally or
    - One where $\vec{E}$ is constant in magnitude.
    - One where we can calculate its area easily:

$$\Phi_E = E \cos\theta \oint_S dA = E \cos\theta A$$

- The integral is only ever easy if the problem has a high degree of symmetry.

Recall: Field outside of sphere of radius, R, carrying charge Q

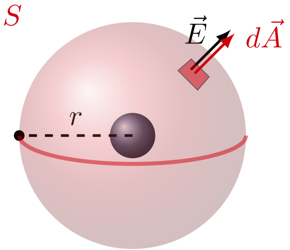
Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $4\pi r^2 E(r)$
3. Determine the charge enclosed: $Q^{enc} = Q$
4. Put it all together:

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

$$4\pi r^2 E(r) = \frac{Q}{\epsilon_0}$$

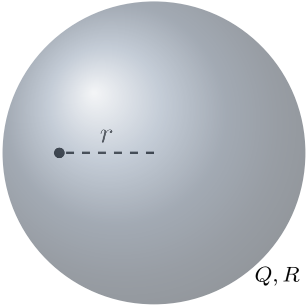
$$\therefore \boxed{E(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}}$$



Gauss' Law to find field outside of a spherical charge of radius, R, and charge, Q.

Field *inside* of a uniform sphere of radius, R, and charge, Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

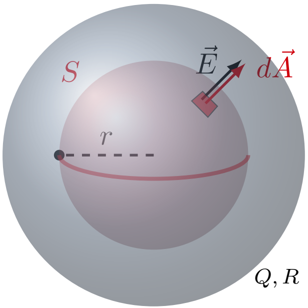


Gauss' Law to find field inside of a spherical charge, Q, of radius.

Field *inside* of a uniform sphere of radius, R, and charge, Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius $r < R$.



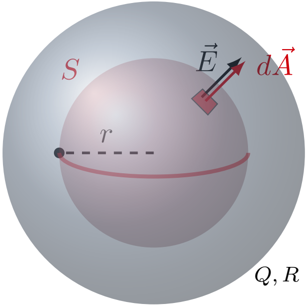
Gauss' Law to find field inside of a spherical charge, Q, of radius.

Field *inside* of a uniform sphere of radius, R, and charge, Q

Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius $r < R$.
2. Evaluate the flux integral (same as before):

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \oint E(r) dA = E(r) \oint dA$$
$$= 4\pi r^2 E(r)$$



Gauss' Law to find field inside of a spherical charge, Q, of radius.

Field *inside* of a uniform sphere of radius, R, and charge, Q

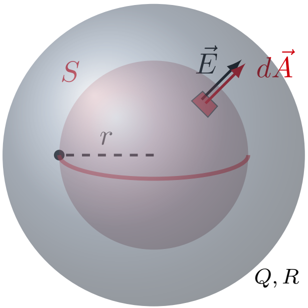
Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius $r < R$.
2. Evaluate the flux integral: $4\pi r^2 E(r)$
3. Determine the charge enclosed. Use the charge density, $\rho = \frac{Q}{V}$:

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$Q^{enc} = \rho \frac{4}{3}\pi r^3 = \frac{Q}{\frac{4}{3}\pi R^3}\frac{4}{3}\pi r^3$$

$$= Q\frac{r^3}{R^3}$$



Gauss' Law to find field inside of a spherical charge, Q, of radius.

Field *inside* of a uniform sphere of radius, R, and charge, Q

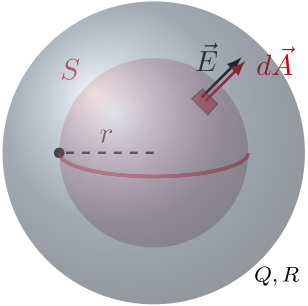
Want the field, $\vec{E}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius $r < R$.
2. Evaluate the flux integral: $-4\pi r^2 g(r)$
3. Determine the mass enclosed: $Q^{enc} = Q\frac{r^3}{R^3}$
4. Put it all together:

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

$$4\pi r^2 E(r) = \frac{Q}{\epsilon_0}\frac{r^3}{R^3}$$

$$\therefore E(r) = \frac{Q}{4\pi\epsilon_0 R^3} r$$



Gauss' Law to find field inside of a spherical charge, Q, of radius.

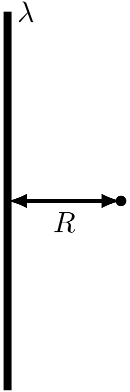
Inside the sphere, the field increases linearly with radius. Outside it decreases with r^2.

Field near an infinitely long charged wire

We want to know the electric field at a distance, r, from an infinitely long wire carrying charge per unit length, λ. We know how to do this with an integral (lecture 45):

$$E(r) = \frac{\lambda r}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{dy}{(y^2 + r^2)^{\frac{3}{2}}}$$

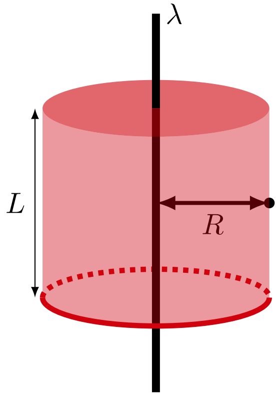
... actually not too bad if you use θ instead of y (trig substitution)!



An infinite wire carrying uniform charge per unit length, λ.

Field near an infinitely long charged wire

We choose a cylindrical gaussian surface of length, L, and radius, r, such that it passes through the point where we want to know the electric field.



An infinite wire carrying uniform charge per unit length, λ.

Field near an infinitely long charged wire

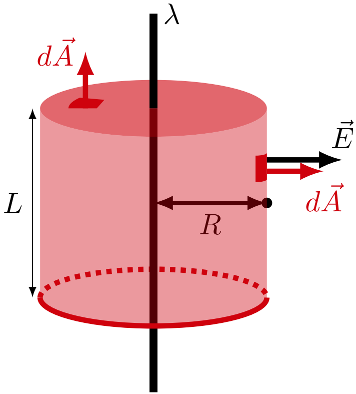
1 Choose a cylindrical gaussian surface.
2 Calculate flux through the surface:
 - Break the surface into:
   1. Endcaps:

$$\Phi_{endcaps} = 0$$

   There are no field lines crossing the endcap.
   2. Wrap-around:

$$\Phi_{wraparound} = \oint_S \vec{E} \cdot d\vec{A} = E \oint_S dA = E(2\pi r L)$$

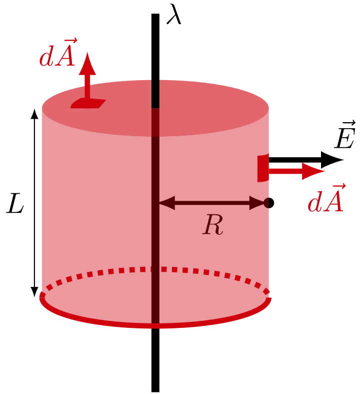
   You can "unwrap" into a rectangle of height L and length $2\pi R$.



An infinite wire carrying uniform charge per unit length, λ.

Field near an infinitely long charged wire

1. Choose a cylindrical gaussian surface.
2. Calculate flux through the surface:
   $\Phi = E(2\pi r L)$
3. Calculate the charge enclosed: $Q^{enc} = \lambda L$



An infinite wire carrying uniform charge per unit length, λ.

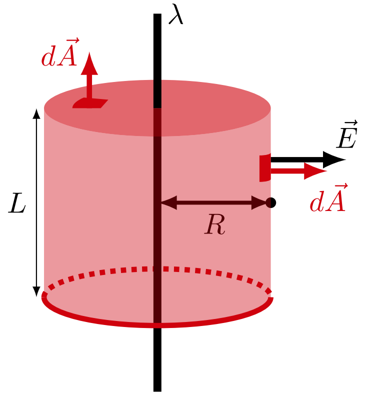
Field near an infinitely long charged wire

1. Choose a cylindrical gaussian surface.
2. Calculate flux through the surface:
   $\Phi = E(2\pi r L)$
3. Calculate the charge enclosed: $Q^{enc} = \lambda L$
4. Putting it all together:

$$\oint \vec{E}(\vec{r}) \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

$$E(2\pi r L) = \frac{\lambda L}{\epsilon_0} \quad \text{The } L \text{ cancels!}$$

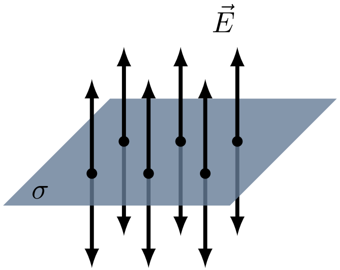
$$\therefore E(r) = \frac{\lambda}{2\pi\epsilon_0 R}$$



An infinite wire carrying uniform charge per unit length, λ.

Infinite plane of charge

We want to know the electric field near an infinite plane carrying uniform charge per unit area, σ

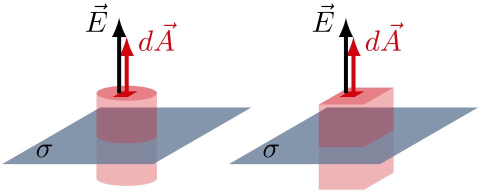


An infinite plane of charge, carrying charge per unit area, σ.

Infinite plane of charge

Two “obvious” choices of gaussian surface:

- Cylinder
- Rectangular prism



Two options for gaussian surfaces to model the infinite plane.

Infinite plane of charge

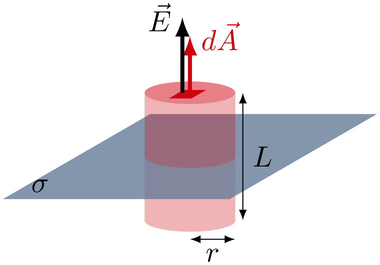
Choosing the cylinder, we have:

- Flux is only through endcaps (there are two of them!):
  $\Phi_E = \Phi_{endcaps} + \Phi_{wraparound} = \mathbf{2}E\pi r^2$
- Charge enclosed by the surface:
  $Q^{enc} = \sigma\pi r^2$

All together:

$$\Phi_E = \frac{Q^{enc}}{\epsilon_0} \rightarrow \quad 2E\pi r^2 = \frac{\sigma\pi r^2}{\epsilon_0}$$
$$\therefore E = \frac{\sigma}{2\epsilon_0}$$

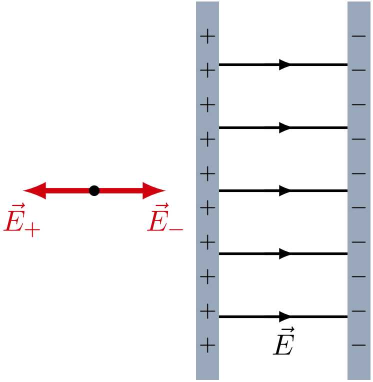
The electric field does not depend on distance from the plane, it's constant everywhere in space!



Using a cylindrical gaussian surface to model the infinite plane.

Field outside of a parallel plate capacitor is zero

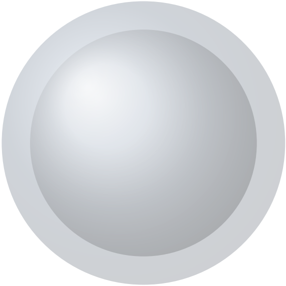
- Two parallel plates carry equal and opposite charge (a capacitor).
- The field does not depend on the distance from either plate.
- The fields from the two plates are in opposite directions everywhere outside the plates (they cancel).
- Between the plates, the field has a constant magnitude.
- Are the charges really in the middle of the plates as illustrated?



Two oppositely-charged planes form a "capacitor", a device that can store energy.

Charges in a conductor: Spherical shell

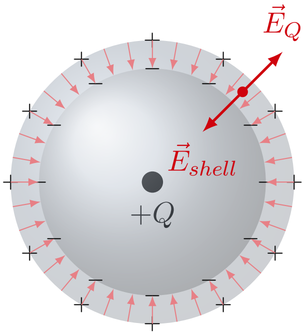
Consider a neutral spherical shell, when we place a charge Q at its centre.



A neutral metallic spherical shell.

Charges in a conductor: Spherical shell

- **The electric field inside of a conductor is always 0**, because the charges are free to move in a conductor.
- If there is a field, charges will move until they have arranged themselves to not move anymore (this usually means that there is a separation of charge, leading to a net electric field of 0).



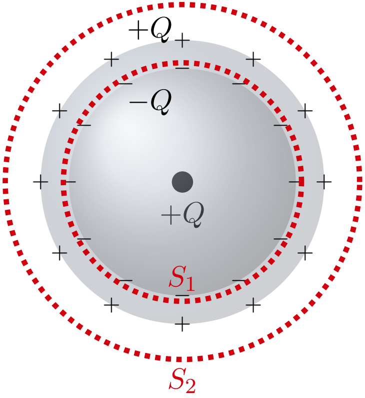
Charges in a (neutral) conducting spherical shell will re-arrange themselves if a charge $+Q$ is placed at its centre (induction), leading to their own electric field.

Charges in a conductor: Spherical shell

We can use Gauss' Law to examine how charges are distributed inside conducting spherical shell with a charge $+Q$ placed at the centre:

- A negative charge, $-Q$, is induced on the inner surface of the conductor. There can be no flux through S_1 (thus no enclosed charge) because S_1 is inside a conductor, where the field must be zero.
- A positive charge, $+Q$ is induced on the outer surface, since the shell is overall neutral.
- The net charge enclosed by S_2 is: $(+Q) + (-Q) + (+Q) = +Q$ (as expected).

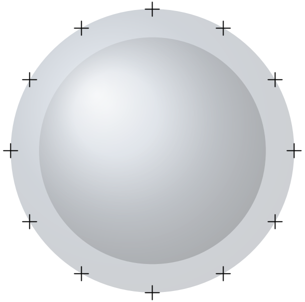
Outside the spherical shell, the electric field is the same as it would be without the shell present!



Gauss' Law implies that the total charge induced on the inner and outer surfaces of the shell must have the same magnitude as the charge placed in the centre.

Charges *on* a conductor

There can be no charge inside a conductor. Thus, if a conductor is charged, all the charges must be on the surface (otherwise, a closed surface in the conductor would enclose charge, thus have a net flux, thus have an electric field across the surface).



On a charged conductor, such as a spherical shell, charges must be on the surface.